

SEMESTER ONE - WEEK 8 – TOPIC 8

CONSTRUCTING THE RESEARCH AIM AND RESEARCH QUESTIONS

CONSTRUCTING THE RESEARCH AIM AND RESEARCH QUESTIONS

- This lecture will show you how to write strong research aims and research questions.
- Your aim and objectives act as the blueprint for
 - what you will study,
 - how you will study it, and
 - how you will report results.

A. WHAT IS THE MAIN RESEARCH AIM (MAIN OBJECTIVE)?

In Kumar's approach, objectives are the goals you set out to attain in your study.

The main aim is an overall statement of the thrust of your study and the main associations or relationships you want to discover or establish.

Example: Main Aim

To design an AI-based solution for crime prevention in the Western Cape, South Africa, by identifying the key crime-prevention requirements and determining the functional features needed for effective implementation.

B. WHAT ARE OBJECTIVES?

Objectives specify the particular aspects of the topic you want to investigate within the main framework of the study.

Objectives are typically listed numerically, and each should focus on a single aspect of the study.

The main aim gives the overall thrust of the study.

Each objective focuses on one aspect of the aim.

B. A THREE-COURSE MEAL METAPHOR?

Think of your research questions as a three-course meal: The main question is the full dining experience. while the sub-questions must be sequenced courses that build cumulative logic.

The starter diagnoses what
is happening,

the main course explains why,

and the dessert delivers
the refined solution,
avoiding overlap or
repetition.

B. A THREE-COURSE MEAL METAPHOR?

How can a framework for integrating generative AI into IT curricula in South African higher education be developed?

How are IT educators in South African higher education currently integrating GenAI, and what inclusion-related challenges emerge in these practices?

What factors enable or constrain inclusive GenAI use in IT in South African higher education?

What strategies and guidelines can be developed to support the integration of GenAI into IT curricula in South African higher education?

EXAMPLE OBJECTIVES

Example: Main Aim

To design an AI-based solution for crime prevention in the Western Cape, South Africa, by identifying the key crime-prevention requirements and determining the functional features needed for effective implementation.

Example: Objectives

- 1. To identify the main crime-prevention challenges and user requirements that an AI-based solution should address in the Western Cape context.*
- 2. To determine the functional features and design considerations required for an AI-based crime-prevention solution to be practical, ethical, and usable in the Western Cape.*

C. AIM, OBJECTIVES, RESEARCH QUESTIONS, AND HYPOTHESES: HOW THEY FIT TOGETHER

1

Which comes first, aims or objectives?

Your aim is your guiding light, shaping the core of your research journey!

2

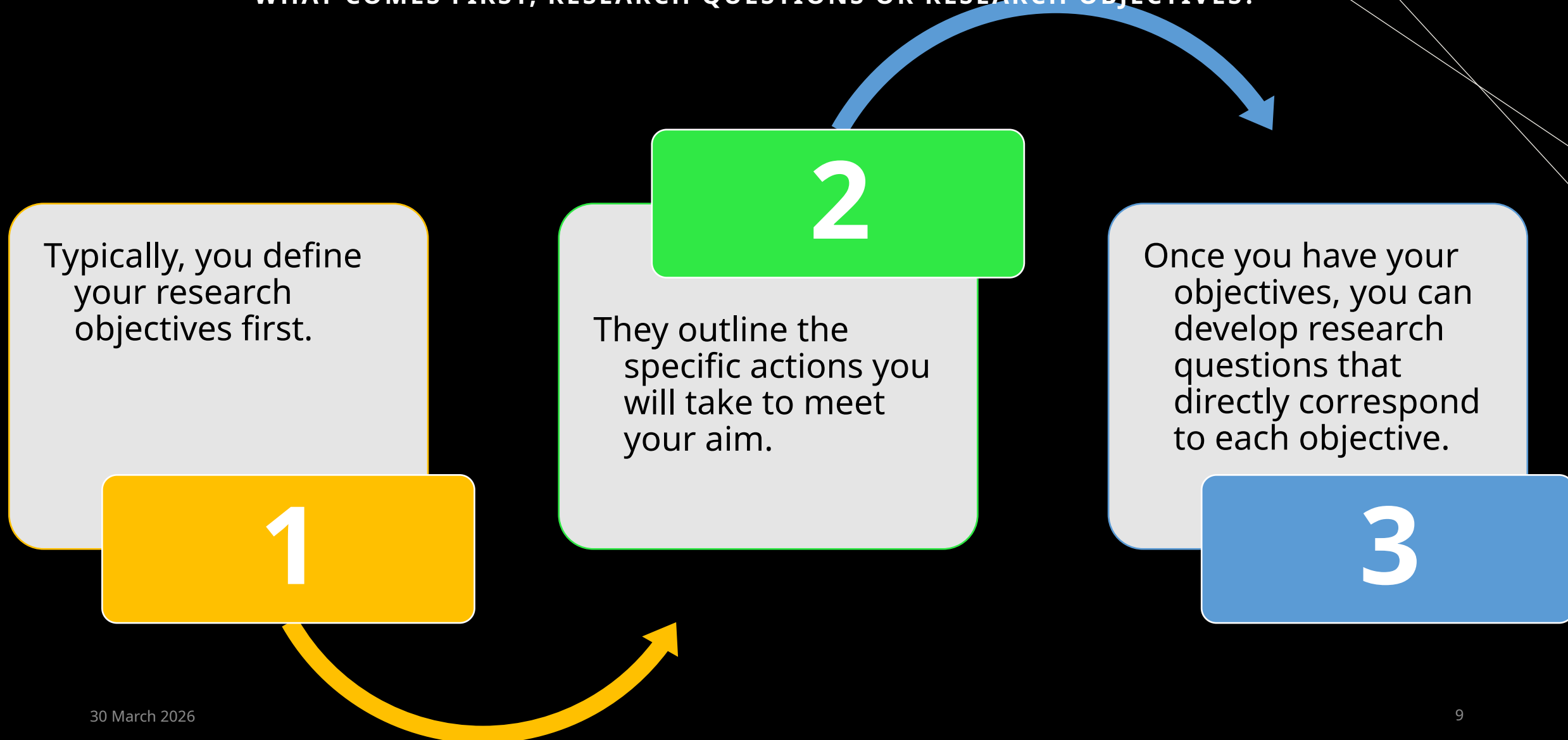
With well-defined objectives, you'll tackle each step with clarity.

3

So, remember: first, embrace your aim, and let your objectives lead the way!

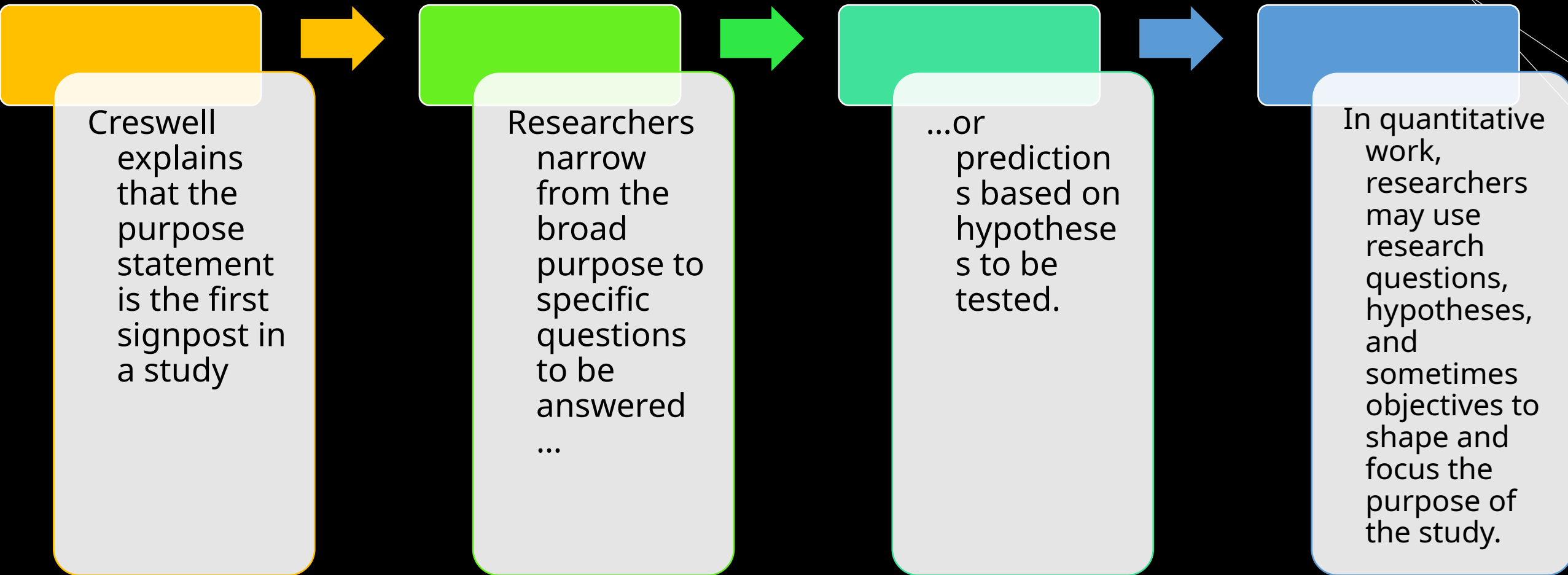
C. AIM, OBJECTIVES, RESEARCH QUESTIONS, AND HYPOTHESES: HOW THEY FIT TOGETHER

WHAT COMES FIRST, RESEARCH QUESTIONS OR RESEARCH OBJECTIVES?



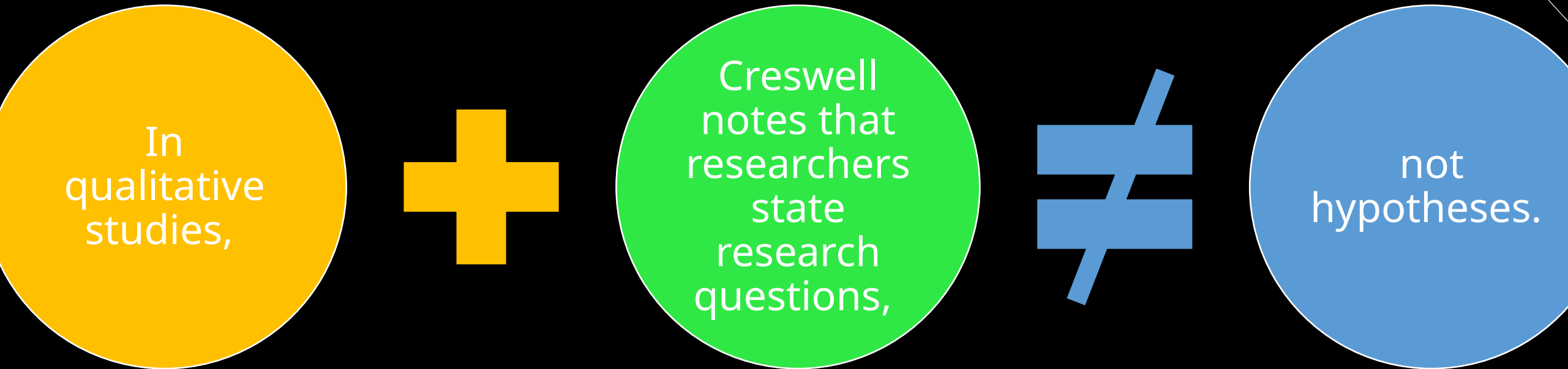
C. AIM, OBJECTIVES, RESEARCH QUESTIONS, AND HYPOTHESES: HOW THEY FIT TOGETHER

WHAT COMES FIRST, RESEARCH QUESTIONS OR RESEARCH OBJECTIVES?



C. AIM, OBJECTIVES, RESEARCH QUESTIONS, AND HYPOTHESES: HOW THEY FIT TOGETHER

IMPORTANT NAMING CONVENTIONS:



**D. STEP-BY-STEP METHOD FOR
CONSTRUCTING A MAIN AIM
AND OBJECTIVES (KUMAR)**

- Kumar proposes raising research questions about what you want to find out,
- and then formulating objectives that grow out of those questions.
- The difference is mainly in how they are written:
- research questions are phrased as questions,
- while objectives transform those questions into behavioural aims using action-oriented wording.

**D. STEP-BY-STEP METHOD FOR
CONSTRUCTING A MAIN AIM
AND OBJECTIVES (KUMAR)**

Use this sequence in your own work:

1. Clarify the topic and the setting (domain, organisation, community, system, or platform).
2. Raise research questions: list everything you want to find out about the topic.
3. Group and reduce: remove overlapping questions, combine similar ones, and keep what is feasible.
4. Write the main aim: a single sentence that captures the overall thrust of the study.
5. Write objectives: convert each key research question into an objective using action verbs.
6. Check feasibility: assess whether the objectives can be achieved within the time, access, resources, and expertise you have.

E. WORDING RULES AND QUALITY CHECKS FOR OBJECTIVES (KUMAR)

- Kumar emphasises that research objectives must be clear, specific, and unambiguous.
- Research objectives should be listed under two headings: **Main Aim** and **Objectives**.
- Objectives should
 - be numerically listed,
 - each should cover only one aspect, and
 - you should use action-oriented verbs such as “to determine”, “to identify”, “to ascertain”, “to measure”, and “to explore”.

**E. WORDING RULES
AND QUALITY
CHECKS FOR
OBJECTIVES (KUMAR)**

Be careful:

Kumar states that the way you word the main aim and objectives determines how the research is classified (for example, descriptive, correlational, or experimental), and therefore the type of research design you need.

QUICK OBJECTIVE CHECKLIST:

1 Clear:
a reader
understands exactly
what you
will
achieve.



**2 Single
focus:**
One
objective,
one issue.



**3 Action
verb:**
starts with
an action-
oriented
verb.



**4 Specific
scope:**
identifies
what,
who, and
where
(and when
if needed).



**5 Design
aligned:**
wording
matches
your
intended
design
(describe,
relate,
compare,
test).

F. HOW CRESWELL'S CHAPTER SUPPORTS YOUR AIM AND OBJECTIVES

- Creswell highlights that research questions and hypotheses narrow the purpose statement and become major signposts.
- For quantitative studies, the author distinguishes between research questions (inquiry about relationships among variables) and
- hypotheses (predictions about expected relationships or group differences),
- and notes that researchers sometimes use objectives to indicate study goals.

F. HOW CRESWELL'S CHAPTER SUPPORTS YOUR AIM AND OBJECTIVES

Useful Creswell guidelines you can apply when you draft objectives:

- Keep variables visible in quantitative framing: questions and hypotheses include variables that are described, related, or compared.
- Use a consistent word order to make key variables easy to see, usually stating the independent variable first and the dependent variable second.
- Consider organising quantitative work from descriptive to inferential: describe key variables first, then ask questions that relate variables or compare groups.

**Example: Applying
Creswell's guidelines
to a quantitative
study**

Topic: Mobile banking adoption among elderly users in the Western Cape

Main aim:

To examine how digital literacy influences mobile banking adoption among elderly users in the Western Cape.

Why this fits Creswell's guidance:

1. The independent variable is stated first: **digital literacy**.
2. The dependent variable is stated second: **mobile banking adoption**.
3. The wording keeps the variables visible and easy to track.

**Example: Applying
Creswell's guidelines
to a quantitative
study.**

**Sub-objectives are organised from
descriptive to inferential**

Descriptive objective 1

To describe the level of digital literacy among elderly mobile banking users in the Western Cape.

Descriptive objective 2

To describe the level of mobile banking adoption among elderly users in the Western Cape.

Inferential objective 3

To examine the relationship between digital literacy and mobile banking adoption among elderly users in the Western Cape.

This follows Creswell's recommendation to move from **describing variables first** to **relating variables afterwards**.

Matching research questions:

1. What is the level of digital literacy among elderly users in the Western Cape?
2. What is the level of mobile banking adoption among elderly users in the Western Cape?
3. What is the relationship between digital literacy and mobile banking adoption among elderly users in the Western Cape?

Example: Applying Creswell's guidelines to a quantitative study.

**Example: Applying
Creswell's guidelines
to a quantitative
study.**

- Creswell helps us make objectives clearer by telling us to keep variables visible,
- keep the wording consistent, and
- organise quantitative work from descriptive to inferential.
- In this example, we first describe **digital literacy** and **mobile banking adoption** separately, and then examine their relationship.

1) Main research aim template

To [action verb] [central phenomenon, relationship, or difference] among/between [participants, systems, or cases] in/at [site or context], in order to [intended contribution or outcome].

G. TEMPLATES YOU CAN REUSE

G. TEMPLATES YOU CAN REUSE

2) Objective templates

- **Descriptive:** To describe [variable/phenomenon] among [population] in [context].
- **Relationship:** To examine the relationship between [independent variable] and [dependent variable] among [population] in [context].
- **Comparison:** To compare [group A] and [group B] on [dependent variable] in [context].
- **Qualitative exploration:** To explore how/why [participants] experience/understand [phenomenon] in [context].

H. AN ICT EXAMPLE (AIM AND OBJECTIVES ALIGNED TO QUESTIONS)

Topic: AI code assistants and code quality in an ICT team.

Main research aim:

To examine how the use of AI code assistants influences code quality in a selected software development team, and to identify practices that support reliable and maintainable outputs.

H. AN ICT EXAMPLE (AIM AND OBJECTIVES ALIGNED TO QUESTIONS)

Possible objectives (quantitative):

1. To determine the extent to which developers in the selected team use AI code assistants in their daily workflow.
2. To measure post-release defect rates for features developed with and without AI assistant support over a defined period.
3. To examine the relationship between AI assistant usage intensity and post-release defect rates, controlling for developer experience level.
4. To compare code review outcomes for AI-assisted commits and non-AI commits on agreed quality indicators.

H. AN ICT EXAMPLE (AIM AND OBJECTIVES ALIGNED TO QUESTIONS)

Possible objectives (qualitative):

1. To explore how developers decide when to rely on AI suggestions and when to reject them.
2. To describe developers' verification and testing practices when AI-generated code is used.
3. To identify perceived benefits, risks, and governance needs related to AI assistant adoption in the team.

- A strong research study requires clear alignment between the research aim, objectives, and research questions.
- This alignment ensures that every objective is investigated through a corresponding research question, maintaining the golden thread from the research problem through data collection and to the findings.

I. LINKING RESEARCH OBJECTIVES AND RESEARCH QUESTIONS (ICT EXAMPLES)

I. LINKING RESEARCH OBJECTIVES AND RESEARCH QUESTIONS (ICT EXAMPLES)

1. Why Alignment Matters

- Research objectives describe what the study intends to achieve, while research questions specify what must be answered in order to achieve those objectives.
- Each objective should be addressed by at least one research question, and each research question should clearly link back to an objective.

**I. LINKING RESEARCH
OBJECTIVES AND
RESEARCH QUESTIONS
(ICT EXAMPLES)**

2. Alignment Rules

- One objective should correspond to one research question.
- Objectives and questions must share the same focus and level of specificity.
- Questions must be researchable through data collection and analysis.

3. MORE ICT EXAMPLES

EXAMPLE 1: CLOUD COMPUTING ADOPTION IN SMES

Main Aim: To examine the factors influencing cloud computing adoption among SMEs in South Africa.

Objective 1: To identify key technological factors influencing the adoption of cloud computing.

Research Question 1: What technological factors influence the adoption of cloud computing among SMEs?

Objective 2: To evaluate the role of organisational ICT readiness on the adoption of cloud computing.

Research Question 2: How does organisational readiness affect SMEs' decisions to adopt cloud computing?

Objective 3: To assess the impact of external environmental factors on the adoption of cloud computing.

Research Question 3: What external environmental factors influence the adoption of cloud computing?

Main Aim: To investigate the impact of cybersecurity awareness training on phishing vulnerability among ICT users in higher education.

Objective 1: To measure changes in phishing recognition rates before and after training.

Research Question 1: To what extent does awareness training improve phishing recognition among ICT users?

Objective 2: To determine the relationship between ICT experience and training outcomes.

Research Question 2: How does prior ICT experience influence the effectiveness of cybersecurity training?

Objective 3: To compare different training delivery methods in reducing phishing susceptibility.

Research Question 3: Which delivery method is more effective in reducing phishing susceptibility?

3. MORE ICT EXAMPLES

EXAMPLE 2: CYBERSECURITY AWARENESS TRAINING

J. SUMMARY

Alignment between objectives and research questions strengthens the coherence, validity, and academic defensibility of a research project.

Well-aligned studies ensure that the research design, data collection, and findings directly address the study's intended purpose.

**K. TUTORIAL QUESTION:
THE GOLDEN THREAD
(5 BLANKS)**

**ANSWER BANK
(6 OPTIONS)**

- A. variables
- B. action verb
- C. feasible
- D. research question
- E. thrust
- F. problem statement

In research, the **main aim** is the overall [_____] of the study, while the **sub-objectives** break that larger purpose into smaller investigable parts. A strong objective should begin with an [_____] such as “to identify” or “to compare,” and it should focus on only one issue so that it remains clear and specific. Each objective should correspond to at least one [_____] so that the study remains logically connected from planning through to findings. In quantitative framing, Creswell recommends keeping the [_____] visible so that relationships or comparisons are clear. Finally, even a well-worded set of objectives must still be [_____] , meaning it can realistically be achieved with the time, access, and resources available to the researcher.

**K. TUTORIAL QUESTION:
THE GOLDEN THREAD
(5 BLANKS)**

**ANSWER BANK
(6 OPTIONS)**

- A. variables
- B. action verb
- C. feasible
- D. research question
- E. thrust
- F. problem statement

In research, the **main aim** is the overall **[thrust]** of the study, while the **objectives** break that larger purpose into smaller investigable parts. A strong objective should begin with an **[action verb]** such as “to identify” or “to compare,” and it should focus on only one issue so that it remains clear and specific. Each objective should correspond to at least one **[research question]** so that the study remains logically connected from planning through to findings. In quantitative framing, Creswell recommends keeping the **[variables]** visible so that relationships or comparisons are clear. Finally, even a well-worded set of objectives must still be **[feasible]**, meaning it can realistically be achieved with the time, access, and resources available to the researcher.

THANK YOU

Q&A